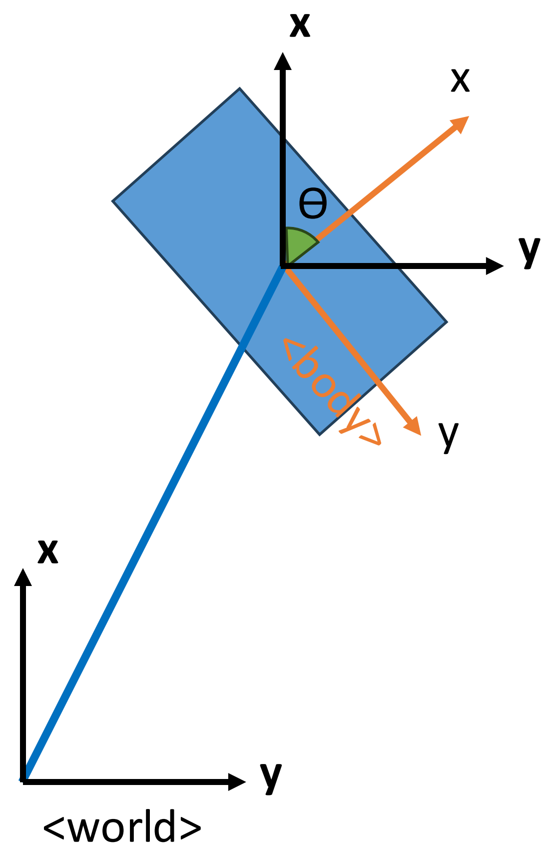
# CLMVS – lab4
# ROV with 3 dofs




$$m\ddot{x} = -\alpha\dot{x} + F_X$$
$$m\ddot{y} = -\beta\dot{y} + F_Y$$
$$J\dot{w} = -\gamma w + N_Z$$

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx \end{cases}$$

$$\frac{d}{dt}x = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\alpha/m & 0 & 0 \\ 0 & 0 & 0 & 0 & -\beta/m & 0 \\ 0 & 0 & 0 & 0 & 0 & -\gamma/J \end{pmatrix} \begin{bmatrix} r_x \\ r_y \\ \theta \\ \dot{r_x} \\ \dot{r_y} \\ w \end{bmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1/m & 0 & 0 \\ 0 & 1/m & 0 \\ 0 & 0 & 1/J \end{pmatrix} \begin{bmatrix} F_x \\ F_y \\ N_z \end{bmatrix}$$

# Computing Kc

We should choose $Kc_{surge}$ such that $\left(A + BKc_{surge}\right)$ is assymptotically stable.

$$A + BK_{surge} = \begin{pmatrix} 0 & 1 \\ 0 & -\frac{\alpha}{m} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \frac{k_{c_1}}{m} & \frac{k_{c_4}}{m} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \frac{k_{c_1}}{m} & -\frac{\alpha}{m} + \frac{k_{c_4}}{m} \end{pmatrix}$$

$$\det\left(\lambda I - \left[A + BK_{surge}\right]\right) = \begin{vmatrix} \lambda & -1 \\ -\frac{k_{c_1}}{m} & \lambda + \frac{\alpha}{m} - \frac{k_{c_4}}{m} \end{vmatrix}$$

$$= \lambda\left(\lambda + \frac{\alpha - k_{c_4}}{m}\right) - \frac{k_{c_1}}{m}$$

$$= \lambda^2 + \left(\frac{\alpha - k_{c_4}}{m}\right)\lambda + \left(-\frac{k_{c_1}}{m}\right) = 0$$

# Computing Kc

We should choose $K_{C_{surge}}$ such that $\left(A + BKc_{surge}\right)$ is assyptotically stable.

$$\det\left(\lambda I - \left[A + BK_{C_{surge}}\right]\right) = \lambda^2 + \left(\frac{\alpha - k_{x_{c4}}}{m}\right)\lambda + \left(-\frac{k_{x_{c1}}}{m}\right) = 0$$

$$\begin{cases} -\dfrac{k_{c_1}}{m} = w_n^2 \\ \dfrac{\alpha - k_{c_4}}{m} = 2\zeta w_n \end{cases}$$

Damping coefficient

Natural frequency

For our exercise we can choose $\zeta = 1,$ wn=1 rad/s

$$k_{c_1} = -m w_n^2$$

$$k_{c_4} = \alpha - 2m\zeta w_n$$

# Computing Kc

We repeat the same method for the other 2 components y and theta.

$$K_C = \begin{bmatrix} k_{c_1} & 0 & 0 & k_{c_4} & 0 & 0 \\ 0 & k_{c_2} & 0 & 0 & k_{c_5} & 0 \\ 0 & 0 & k_{c_3} & 0 & 0 & k_{c_6} \end{bmatrix}$$

$$surge: \begin{cases} k_{c_1} = -mw_n^2 \\ k_{c_4} = \alpha - 2m\zeta w_n \end{cases}$$

$$sway: \begin{cases} k_{c_2} = -mw_n^2 \\ k_{c_5} = \beta - 2m\zeta w_n \end{cases}$$

$$yaw: \begin{cases} k_{c_3} = -J_w w_n^2 \\ k_{c_6} = \gamma - 2J_w\zeta w_n \end{cases}$$

# Recall: Computing Ko

- $K_o = \begin{bmatrix} k_{x_1} & 0 & 0 \\ 0 & k_{x_2} & 0 \\ 0 & 0 & k_{x_3} \\ k_{x_4} & 0 & 0 \\ 0 & k_{x_5} & 0 \\ 0 & 0 & k_{x_6} \end{bmatrix}$

$$surge: \begin{cases} k_{x_1}\dfrac{\alpha}{m} + k_{x_4} = w_n^2 \\ \dfrac{\alpha}{m} + k_{x_1} = 2\zeta w_n \end{cases}$$

$$sway: \begin{cases} k_{x_2}\dfrac{\beta}{m} + k_{x_5} = w_n^2 \\ \dfrac{\beta}{m} + k_{x_2} = 2\zeta w_n \end{cases}$$

$$yaw: \begin{cases} k_{x_3}\dfrac{\gamma}{J_w} + k_{x_6} = w_n^2 \\ \dfrac{\gamma}{J_w} + k_{x_3} = 2\zeta w_n \end{cases}$$

Tune ζ, w so that the observer converges faster than the controller

**Exercise**

1. Starting from the given template lab4_ex1.m, implement a closed loop control, assuming that state is fully measurable, $x_0$ = [0;0;0;0;0;0], target state is $x_f$ = [-1;2;3;0;0;0], in time T = 20 s. Properly tune the control gain matrix $K_c$. Plot the state evolution.

2. Implement a Luenberger observer, assuming that the state is observable, being the pose variables only measured, tuning the observer gain matrix $K_o$. Assume that the initial state is known with error [1, -1, 1.5, 0.5, -0.5, -1.5]. Plot the difference between the true and estimated state.

3. Now use the Luenberger observer to close the control loop, using the estimated state in the control feedback. Plot the state evolution.

4. Repeat 2, 3 assuming that the whole state is measured.